REPORT ON ARTIFICIAL INTELLIGENCE

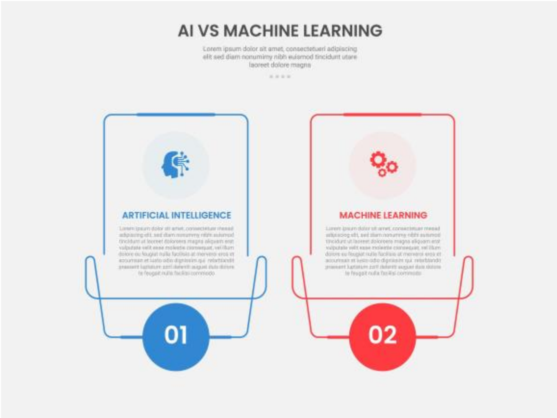
Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning from data, understanding language, recognizing images, solving problems, making decisions, and generating content. AI systems use algorithms and large amounts of data to identify patterns and improve their performance over time.

Today, AI is used in many applications such as virtual assistants, recommendation systems, self-driving vehicles, healthcare diagnosis, fraud detection, and content generation.

Q2. Difference Between AI, Machine Learning, Generative AI, and Agentic AI

Technology	Description	Example
Artificial Intelligence (AI)	The broad field of creating machines that can mimic human intelligence.	Chess-playing software
Machine Learning (ML)	A subset of AI where systems learn patterns from data without being explicitly programmed for every task.	Email spam detection
Generative AI	A type of AI that creates new content such as text, images, audio, or code.	ChatGPT generating an essay
Agentic AI	AI systems that can plan, make decisions, and perform actions autonomously to achieve goals.	An AI assistant that schedules meetings and sends emails automatically



Q3. Explain Any Three Open-Source AI Models Available on Hugging Face

1. Llama 3

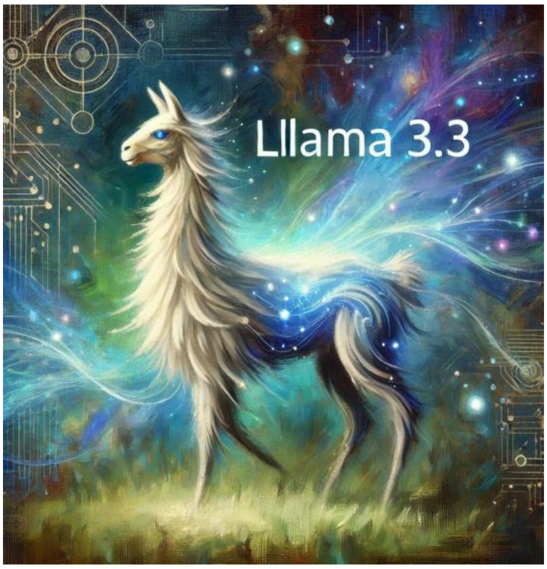
Llama 3 is a family of open-source large language models developed by Meta. It is designed for tasks such as text generation, question answering, summarization, and coding assistance. It offers strong performance while remaining accessible to researchers and developers.

2. Mistral 7B

Mistral 7B is a lightweight yet powerful language model created by Mistral AI. Despite having fewer parameters than many larger models, it provides excellent performance in reasoning, text generation, and conversational applications.

3. Falcon 7B

Falcon 7B is an open-source language model developed by the Technology Innovation Institute (TII). It is widely used for natural language processing tasks and is known for its efficiency and strong performance in text understanding and generation.



Q4. What Is the Difference Between Traditional Programming and AI-Assisted Development?

Traditional Programming

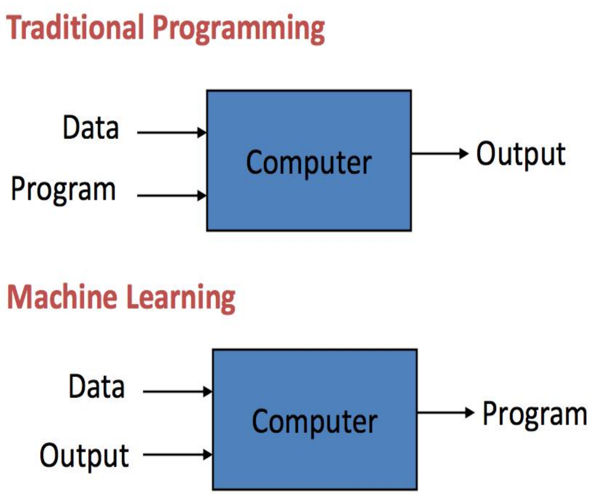
In traditional programming, developers write explicit rules and instructions that the computer follows. The output depends entirely on the logic written by the programmer.

Example:
If a programmer wants a system to calculate discounts, they must write all the conditions and formulas manually.

AI-Assisted Development

In AI-assisted development, AI tools help developers write code, suggest solutions, generate documentation, identify bugs, and automate repetitive tasks. The developer still guides the process, but AI increases productivity and efficiency.

Example:
A developer can ask an AI coding assistant to generate a function for calculating discounts, and the AI can produce a working code example.



Q5. Explain AI Hallucinations with One Example

AI hallucination occurs when an AI system generates information that appears correct and convincing but is actually false, inaccurate, or unsupported by facts.

Hallucinations happen because AI models predict likely responses based on patterns in training data rather than verifying information from reliable sources.

Example: If an AI states that a person won an award they never received, that information is a hallucination because it is not based on real facts.

How generative AI hallucinations work